

✓ greeting() - Return a string

```
def greeting():  
    a = "Good Morning"  
    return a  
  
print(greeting())  
print(type(greeting()))
```

```
Good Morning  
<class 'str'>
```

✓ summoftthree() - Sum of 3 inputs

```
def summoftthree():  
    a,b,c = int(input()), int(input()), int(input())  
    return a+b+c  
  
print(summoftthree())
```

```
13  
15  
7  
35
```

✓ fact(n) - Factorial (Recursive)

```
def fact(n):  
    if n == 0:  
        return 1  
    else:  
        return n * fact(n-1)  
  
n = int(input("Enter a num for fact: "))  
a = fact(n)  
print(f"{n}! = {a}")
```

```
Enter a num for fact: 13  
13! = 6227020800
```

✓ sumofnum() - Sum of numbers from input

```
def sumofnum():  
    a = list(map(int, input().split()))  
    return sum(a)  
print(sumofnum())
```

```
1 2 3  
6
```

✓ calculate_of_three(a,b,c) - Addition, Subtraction, Multiplication, Power, Division

```
def calculate_of_three(a, b, c):
    addition = a + b + c
    subtraction = a - b - c
    multiplication = a * b * c
    power = a ** b
    div = a / b

    return addition, subtraction, multiplication, power, div

a, b, c = map(int, input().split())
print(calculate_of_three(a, b, c))
```

```
13 15 7
(35, -9, 1365, 51185893014090757, 0.8666666666666667)
```

✓ Is primenumber(n) - Is Prime or Not

```
def isprimenumber(n):
    for i in range(2, n):
        if n % i == 0:
            return 0
    return 1
a = int(input("Enter a num: "))
print(isprimenumber(a))
```

```
Enter a num: 13
1
```

✓ prime in a range(m,n) - Primes in a Range (returns list)

python

```
def primeinarange(m, n):
    a = []
    for i in range(m, n+1):
        if i > 1:
            for j in range(2, i):
                if i % j == 0:
                    break
            else:
                a.append(i)
    return a

print(sum(primeinarange(2, 10)))
print(primeinarange(2, 10))
```

```
17
[2, 3, 5, 7]
```

✓ even odd in a range(a,b) - Even and Odd lists in a Range

```
def evenoddinarange(a,b):
    even = []
    odd = []
    for i in range(a, b+1):
        if i % 2 == 0:
            even.append(i)
        else:
            odd.append(i)
    return even, odd

print("Even List:")
print(evenoddinarange(1, 30)[0])
print("Odd List:")
print(evenoddinarange(1, 30)[1])
```

```
Even List:
[2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30]
Odd List:
[1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29]
```

✓ expo(a,b) - Exponentiation without ** operator

```
def expo(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    ans = 1
    for i in range(b):
        ans = ans * a
    return ans

print(expo(a,b))
```

```
Enter a num: 13
Enter a num: 15
51185893014090757
```

✓ Sliding Window Sum (k=5)

```
a = list(map(int, input().split()))
k = 5

for i in range(len(a) - k + 1):
    w_sum = 0
    for j in range(i, i + k):
        w_sum += a[j]
    print(w_sum)
```

```
13 7
```

✓ has pair(arr,t) - Two Sum / Has Pair with given sum

```
def haspair(arr,t):
    left=0
    right=len(arr)-1
    while left<right:
        s=arr[left]+arr[right]
        if s==t:
            return True
        elif s<t:
            left+=1
        else:
            right-=1
    return False
print(haspair([1,2,3,4,5,6],10))
```

```
True
```

✓ sq and sqroot(n) - Square and Square Root

```
def sqandsqroot(n):
    sq = n ** 0.5
    sqroot = n ** n
    return sq, sqroot

x = int(input("Enter a num: "))
print(sqandsqroot(x))
```

```
Enter a num: 7
(2.6457513110645907, 823543)
```

✓ first(a) - Return input as-is

```
def first(a):
    a = input()
    return a

print(first(a))
```

```
Hani
Hani
```

▼ area of triangle(l,b) - Area of Triangle

```
def areaoftriangle(l,b):
    area = 0.5 * l * b
    return area

l = int(input("Enter a num for length: "))
b = int(input("Enter a num for breadth: "))
print(areaoftriangle(l,b))
```

```
Enter a num for length: 13
Enter a num for breadth: 15
97.5
```

▼ linear_search(a, target) - Linear Search

```
def linear_search(a, target):
    for i in range(len(a)):
        if a[i] == target:
            return f"Found at {i}"
    return "Not Found"

a = list(map(int, input().split()))
target = int(input("Enter a num: "))
print(linear_search(a, target))
```

```
1 3 1 5 7
Enter a num: 7
Found at 4
```

▼ quad eq(a,b,c) - Quadratic Equation Discriminant

```
def quadeq(a,b,c):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    c = int(input("Enter a num: "))
    d = (b**2) - (4*a*c)
    if d > 0:
        print("Roots are real and unequal")
    elif d == 0:
        print("Roots are real and equal")
    else:
        print("Roots are imaginary")
    return d

print(quadeq(5,6,7))
```

```
Enter a num: 7
Enter a num: 13
Enter a num: 15
Roots are imaginary
-251
```

▼ swap with c(a,b) - Swap using temp variable

```
def swapwithc(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
```

```

c = a
a = b
b = c
return a, b

print(swapwithc(a,b))

```

```

Enter a num: 13
Enter a num: 15
(15, 13)

```

✓ swap without c(a,b) - Swap without temp variable

```

def swapwithoutc(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    a, b = b, a
    return a, b

print(swapwithoutc(a,b))

```

```

Enter a num: 15
Enter a num: 13
(13, 15)

```

✓ swap(a,b) - Swap using arithmetic

```

def swap(a,b):
    a = int(input("Enter a num: "))
    b = int(input("Enter a num: "))
    a = a + b
    b = a - b
    a = a - b
    return a, b

print(swap(a,b))

```

```

Enter a num: 13
Enter a num: 7
(7, 13)

```

✓ random gen(a) - Random Number Generator

```

import random as rd
def randomgen(a):
    a = rd.randint(0,10)
    return a
print(randomgen(a))

```

```

3

```

✓ distance(d) - KM to Miles

```

#1 km = 0.6 miles
def distance(d):
    d = 0.6 * d
    return d

n = float(input())
print(distance(n))

```

```

13
7.8

```

✓ distance(d) - Miles to KM

```
#1mile = 1.6 kms
def distance(d):
    d = 1.6 * d
    return d

n = float(input())
print(distance(n))
```

```
7
11.200000000000001
```

✓ cel to fah(t) - Celsius to Fahrenheit

```
def celtofah(t):
    t = (9/5) * t + 32
    return t
print(celftofah(50))
```

```
122.0
```

✓ fah to cel(t) - Fahrenheit to Celsius

```
def fahto cel(t):
    t = (5/9) * (t - 32)
    return t
print(fahto cel(122))
```

```
50.0
```

✓ sign(n) - Positive / Negative / Zero

```
def sign(n):
    if n > 0:
        print("Positive")
    elif n < 0:
        print("Negative")
    else:
        print("Zero")

m = int(input())
sign(m)
```

```
13
Positive
```

✓ even odd(m) - Even or Odd with type check

```
def evenodd(m):
    if type(m) == int:
        if m % 2 == 0:
            print("Even")
        else:
            print("Odd")
    else:
        print("Invalid Input")
n = int(input())
evenodd(n)
```

```
13
Odd
```

✓ BinarySearch(a,high,low,target) - Binary Search

```
def BinarySearch(a,high,low,target):
    while low <= high:
        mid = (high + low) // 2
        if a[mid] == target:
            return mid
        elif a[mid] < target:
            low = mid + 1
        else:
            high = mid - 1
    return -1
a = list(map(int, input().split()))
target = int(input())
print(BinarySearch(a,len(a)-1,0,target))
```

```
7
13
-1
```

✓ year(y) - Leap Year Check (version 1 - incomplete)

```
def year(y):
    if y % 4 == 0:
        if y % 100 == 0:
            if y % 400 == 0:
                print("Leap Year")
            else:
                print("Not a Leap Year")
        else:
            print("Leap Year")
    else:
        print("Not a Leap Year")

z = int(input())
year(z)
```

```
2025
Not a Leap Year
```

✓ largest(a,b,c) - Largest of 3 numbers

```
def largest(a,b,c):
    if a > b and a > c:
        print(a)
    elif b > a and b > c:
        print(b)
    else:
        print(c)

x, y, z = map(int, input().split())
largest(x, y, z)
```

```
1 3 7
7
```

✓ largest(a) - Largest in a List

```
def largest(a):
    big = a[0]
    for i in range(1, len(a)):
        if a[i] > big:
            big = a[i]
    return big

a = list(map(int, input().split()))
print(largest(a))
```

```
1 3 7 1 5
7
```

✓ Is prime(p) - Is Prime (version 2)

```
def isprime(p):  
    for i in range(2, n):  
        if n % i == 0:  
            return 0  
    return 1
```

```
n = int(input())  
print(isprime(n))
```

```
13  
1
```

✓ primeinrange(m,n) - Print Primes in Range

```
def primeinrange(m,n):  
    for i in range(m, n+1):  
        if i > 1:  
            for j in range(2, i):  
                if i % j == 0:  
                    break  
            else:  
                print(i)
```

```
primeinrange(2, 10)
```

```
2  
3  
5  
7
```

✓ fact(n) - Factorial version 2

```
def fact(n):  
    if n == 0:  
        return 1  
    else:  
        return n * fact(n-1)
```

```
n = int(input())  
print(fact(n))
```

```
7  
5040
```

✓ multiplication table(m) - Multiplication Table

```
def multiplicationtable(m):  
    for i in range(1, 21):  
        print(f"{m} x {i} = {m * i}")
```

```
n = int(input())  
multiplicationtable(n)
```

```
13  
13 x 1 = 13  
13 x 2 = 26  
13 x 3 = 39  
13 x 4 = 52  
13 x 5 = 65  
13 x 6 = 78  
13 x 7 = 91  
13 x 8 = 104  
13 x 9 = 117  
13 x 10 = 130  
13 x 11 = 143  
13 x 12 = 156  
13 x 13 = 169
```



```

13 x 14 = 182
13 x 15 = 195
13 x 16 = 208
13 x 17 = 221
13 x 18 = 234
13 x 19 = 247
13 x 20 = 260

```

✓ Fibonacci Series (inline)

```

n = 5
first, second = 0, 1

for i in range(n):
    print(first)
    first, second = second, first + second

```

```

0
1
1
2
3

```

✓ fiboserries(n) - Fibonacci Series (function)

```

def fiboserries(n):
    first, second = 0, 1
    for i in range(n):
        print(first)
        first, second = second, first + second

```

```
fiboserries(5)
```

```

0
1
1
2
3

```

✓ fib(n) - Nth Fibonacci (Recursive)

```

def fib(n):
    if n == 0:
        return 0
    if n == 1:
        return 1
    else:
        return fib(n-1) + fib(n-2)

```

```

n = int(input())
print(fib(n))

```

```

7
13

```

✓ sum of n(numbers) - Sum of N numbers

```

def sumofn(numbers):
    total = 0
    numbers = map(int, input().split())
    for i in numbers:
        total += i
    return total

```

```
print(sumofn(a))
```

```

1 3 15 7
26

```

✓ Sum of list (inline)

```
a = [2,3,4]
sum = 0
for i in a:
    sum = sum + i
print(sum)
```

9

✓ power(a,n) - Power

```
def power(a, n):
    ans = a ** n
    return ans

a, n = map(int, input().split())
print(power(a, n))
```

7 13

96889010407

✓ Reverse a string (slice)

```
normal = "123"
reverse = normal[::-1]
print(reverse)
```

321

✓ Reverse digits (while loop)

```
num = 123
while num > 0:
    digit = num % 10
    print(digit, end = "")
    num = num // 10
```

321

✓ reverse() - Reverse digits (function with input)

```
def reverse():
    n = int(input())
    while n > 0:
        digit = n % 10
        print(digit, end = "")
        n = n // 10
    reverse()
```

13

31

✓ search Insert(nums, target) - Search Insert Position

```
def searchInsert(nums, target):
    for i in range(len(nums)):
        if nums[i] >= target:
            return i
    return len(nums)
```

```
nums = list(map(int, input().split()))
target = int(input())
print(searchInsert(nums, target))
```

```
13
15
1
```

✓ Solution.nextGreatestLetter - Next Greatest Letter

```
class Solution:
    def nextGreatestLetter(letters, target):
        for ch in letters:
            if ch > target:
                return ch
        return letters[0]
print(Solution.nextGreatestLetter(letters = ["H", "T"], target = "a"))
```

```
H
```

✓ factorsofn(n) + isprime(n) - Factors and Prime Check

```
def factorsofn(n):
    factors = []
    for i in range(1, n):
        if n % i == 0:
            factors.append(i)
    return factors

def isprime(n):
    if n > 1:
        for j in range(2, n):
            if n % j == 0:
                return False
        return True
    return False

m = int(input())
y = factorsofn(m)
```

```
13
```

✓ Solution.search - Search in Rotated Sorted Array

```
class Solution(object):
    def search(self, nums, target):
        left = 0
        right = len(nums) - 1
        while left <= right:
            mid = (left + right) // 2
            if nums[mid] == target:
                return mid
            if nums[left] <= nums[mid]:
                if nums[left] <= target < nums[mid]:
                    right = mid - 1
                else:
                    left = mid + 1
            else:
                if nums[mid] < target <= nums[right]:
                    left = mid + 1
                else:
                    right = mid - 1
        return -1

sol = Solution()
nums = list(map(int, input().split()))
target = int(input())
print(sol.search(nums, target))
```

```
13
15
-1
```

✓ ord() and chr() - ASCII values

```
a = "c"
print(ord(a))

print(ord("c"))

for i in range(ord("A"), ord("Z") + 1):
    print(i, end = " ")

print()

for i in range(32, 101):
    print(chr(i), end=" ")
```

```
99
99
65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90
! " # $ % & ' ( ) * + , - . / 0 1 2 3 4 5 6 7 8 9 : ; < = > ? @ A B C D E F G H I J K L M N O P Q R S T U V W X Y Z [ \ ] ^ _
```

✓ Bubble Sort (inline)

```
l = list(map(int, input().split()))
for i in range(len(l)):
    for j in range(0, len(l) - 1):
        if l[j] > l[j + 1]:
            l[j], l[j + 1] = l[j + 1], l[j]
print(l)
```

```
13 15 7
[7, 13, 15]
```

✓ alphabet(n) - Vowels and Consonants

```
def alphabet(n):
    vowels = []
    consonants = []
    for i in n:
        if i.lower() in "aeiou" or i.upper() in "AEIOU":
            vowels.append(i)
        else:
            consonants.append(i)
    return vowels, consonants

m = input()
vowels, consonants = alphabet(m)
print("Vowels: ", vowels)
print("Consonants: ", consonants)
```

```
H a n i
Vowels: ['a', 'i']
Consonants: ['H', ' ', ' ', ' ', 'n', ' ', '']
```

✓ Perfect Number Check (inline)

```
n = 6
total = 0

for i in range(1, n):
    if n % i == 0:
        total += i
```

```
if total == n:
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```

Perfect Number

✓ Peak Element (inline)

```
b = list(map(int, input().split()))
a = [0] + b + [0]

peaks = []

for i in range(1, len(a) - 1):
    if a[i] > a[i - 1] and a[i] > a[i + 1]:
        peaks.append(i - 1)

if len(peaks) == 0:
    print("No peaks")
else:
    largest = peaks[0]
    for i in peaks:
        if i > largest:
            largest = i
    print(largest)
```

```
1 3 7
2
```

✓ Solution.findPeakElement - Find Peak Element (Binary Search)

```
class Solution(object):
    def findPeakElement(self, nums):
        left = 0
        right = len(nums) - 1
        while left < right:
            mid = (left + right) // 2
            if nums[mid] > nums[mid + 1]:
                left = mid + 1
            else:
                right = mid
        return left
```

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